

KernelScript: Exploring a Typed Unified-Source Programming Model for eBPF

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Abstract

eBPF has become a standard mechanism for extending Linux kernel behavior in networking, observability, and security. Yet realistic eBPF applications are not single programs: developers coordinate verifier-constrained kernel code, userspace loaders, shared maps, generated skeletons, and sometimes kernel module code for kfuncs. This paper studies KernelScript, a beta domain-specific language that explores whether these pieces can be represented in one type-checked source model and lowered to conventional libbpf-compatible artifacts. We present a design analysis of KernelScript’s unified programming model, then evaluate the public implementation at commit 6f9e6e8. On our test host, KernelScript’s own test suite reports 85 successful suites and 1092 passing tests. Of 44 repository examples, 43 compile from KernelScript source and 41 build into generated C/eBPF projects. Successful examples have median 31 lines of KernelScript source and median 472 lines of generated C and build logic, an 11.3x expansion factor. The results show that the compiler centralizes recurring project structure and expose two current limits: struct_ops skeleton compatibility with the installed libbpf headers, and a map-update microbenchmark where generated code uses more instructions than an idiomatic hand-written baseline.

1 Introduction

eBPF gives Linux developers a controlled way to execute application-specific logic in the kernel [6]. It is now used for packet processing, traffic control, tracing, profiling, security policy, and scheduler experimentation. The programming model remains difficult because the artifact boundary is split: an eBPF program is written under verifier constraints, a userspace loader must open and attach it, maps provide shared state, BTF describes kernel types, and newer interfaces such as kfuncs and struct_ops require close coordination between generated skeleton code and kernel headers.

The usual C/libbpf workflow is powerful but verbose [4]. The developer writes kernel-side C, userspace C, Makefiles, pinning logic, map setup, attachment code, and cleanup paths. Libraries in Rust, Go, or Python improve parts of the workflow [1, 3], but they usually preserve the split between the eBPF program and the controller. Tracing languages such as bpftrace provide a higher-level experience, but intentionally narrow the target domain. The systems question is therefore not whether eBPF needs another surface syntax. The question

is whether a language can encode cross-boundary eBPF application structure in a way that is both safer and still compatible with the existing kernel toolchain.

1.1 Why a Unified Source Model?

A single source file is not automatically better than multiple files. The research value comes from making cross-file invariants visible to the compiler. For example, a map declaration is simultaneously a kernel data structure, a userspace file descriptor lookup, a pinning decision, and a type contract between programs. A program reference is simultaneously a source-level function name, a generated BPF program section, a skeleton field, a file descriptor, and possibly a BPF link. In C/libbpf, these relationships are implicit naming conventions and generated-header dependencies. In KernelScript, they can be first-class language values.

This observation motivates the paper’s main scientific question: can an eBPF DSL reduce accidental complexity without reducing the programmer’s ability to target the real Linux BPF stack? A tracing-only language such as bpftrace [2] can be concise by excluding XDP, TC, struct_ops, and kfunc workflows. A host-language binding can be ergonomic by focusing on userspace loading. KernelScript attempts a harder point in the design space: preserve multiple eBPF program types and still give the compiler enough structure to generate the loader, maps, and build logic.

KernelScript is a recent DSL that makes this question concrete. It lets developers write eBPF entry points, helper functions, userspace control flow, maps, tail-call style delegation, ring buffers, perf events, kfunc declarations, and kernel-module code in one source language. The compiler lowers this source to eBPF C, userspace C, optional module C, a Makefile, and then the normal libbpf and bpftool path can produce objects, skeletons, and binaries. Because KernelScript is explicitly beta software, its most useful research role today is as a prototype for studying the design space rather than as a production-ready replacement for libbpf.

This paper makes three contributions. First, we identify the central scientific contribution of KernelScript as a typed, compiler-checked unified-source model for multi-artifact eBPF applications. Second, we describe how this model organizes programs, maps, lifecycle operations, and kernel extension code so that cross-boundary constraints can be checked before load or attach time. Third, we implement and run a reproducible artifact evaluation that measures test coverage, example buildability, generated-code expansion, fea-

ture coverage, and an attach/detach smoke test. The artifact is designed so that a reviewer can rerun the evaluation without changing the KernelScript source tree: all scripts, logs, derived tables, and paper macros are stored outside the cloned repository.

2 Background and Motivation

An eBPF application normally contains at least two components. The first is a restricted kernel program, compiled for the BPF target and accepted only if the kernel verifier can prove memory safety, bounded execution, and legal helper usage. The second is a userspace coordinator that loads objects, finds maps, sets up BPF links, handles signals, and tears resources down. The two components communicate through maps, ring buffers, perf events, or global data.

This split creates three kinds of complexity. **Lifecycle complexity** arises because operations must occur in a valid order: an object must be opened, loaded, attached, read, and detached with type-appropriate APIs. **Representation complexity** appears when the same concept, such as a map or a typed context field, must be represented in source C, generated skeleton headers, userspace lookup code, and BTF-derived type declarations. **Compatibility complexity** appears because eBPF features evolve quickly: struct_ops, kfuncs, dynptrs, TCX, and scheduler extensions depend on kernel, bpftool, and libbpf versions.

The verifier is essential, but it is intentionally late in the pipeline. A developer may write code, compile it, generate skeletons, link a loader, and only then receive a verifier rejection or an attach-time failure. A DSL can help if it moves some of these checks earlier without hiding the underlying eBPF execution model. For example, a language can know that a function marked @xdp expects an xdp_md context and returns an XDP action; it can model maps as typed global variables; and it can distinguish userspace functions from eBPF functions and kfunc definitions.

3 KernelScript Overview

KernelScript uses attributes to assign functions to execution domains. A function marked @xdp, @tc, @probe, @tracepoint, or @perfevent becomes an eBPF entry point. A function marked @helper is callable from eBPF code. A normal main function is compiled into userspace C. A function marked @kfunc can trigger kernel-module generation. The source therefore contains the application structure that would otherwise be scattered across multiple files.

Listing 1: Minimal shape of a unified-source eBPF application.

```
include "xdp.kh"

@xdp fn pass_all(ctx: *xdp_md) -> xdp_action {
    return XDP_PASS
}
```

```
}

fn main() -> i32 {
    var prog = load(pass_all)
    attach(prog, "lo", 0)
    detach(prog)
    return 0
}
```

Maps are declared as typed globals rather than as separate C sections and userspace lookup code. A KernelScript program can declare a hash map from u32 to a struct, use it in an XDP function, and read or update it in userspace. The compiler lowers the declaration to eBPF map definitions, skeleton references, and helper functions. The same pattern applies to ring buffers and perf events: high-level operations remain explicit in source, but the generated code performs the required libbpf calls.

The language also experiments with lifecycle typing. In the source model, load returns a program value and attach consumes a compatible loaded program and target. The repository contains tests for detach API generation and type errors around program references. This is not a proof of full types-tate soundness, but it shows the research direction: lifecycle ordering can be treated as a language-level property rather than only as a runtime convention.

4 Design Principles

Preserve the kernel toolchain. KernelScript does not replace the kernel verifier, clang, bpftool, or libbpf. Instead it generates C and Makefiles that flow through the same toolchain used by C/libbpf projects. This choice gives the prototype immediate access to BTF, skeleton generation, and existing kernel diagnostics. It also means compatibility failures remain visible and measurable.

Make domains explicit. The attribute system separates userspace, eBPF, helper, and kernel-module code. This matters because the same surface syntax cannot mean the same thing in every domain. Userspace can allocate and print freely; eBPF code must obey stack and helper restrictions; kfunc code targets a module build. A unified source model is useful only if the compiler keeps these domains separate internally.

Move verifier-relevant checks earlier. The compiler performs safety analysis before code generation. In our run, safety_demo.ks is rejected because the compiler estimates 608 bytes of stack use, above the 512-byte eBPF limit. This does not make the verifier redundant, but it reduces the number of late failures that require inspecting verifier logs.

Expose low-level escape hatches. eBPF users often need new helpers, kernel functions, or BTF-specific types before high-level libraries expose them. KernelScript supports includes and extern declarations so that a program can refer to generated header declarations. This is important for a research prototype because the Linux BPF API surface changes faster than a DSL can stabilize.

5 Implementation

The public implementation is an OCaml compiler built with dune. It parses KernelScript source, resolves imports and includes, builds symbol tables, performs multi-program analysis and type checking, runs safety analysis, generates an intermediate representation, and emits C artifacts. For an input `foo.ks`, the generated project normally contains `foo.ebpf.c`, `foo.c`, a `Makefile`, and, when kfuncs are present, `foo.mod.c`. The `Makefile` uses `bpftool` to dump `vmlinux.h`, `clang` to produce a BPF object, `bpftool` to generate a skeleton header, and `gcc` to link the userspace loader with `libbpf`, `libelf`, and `zlib`.

The compiler’s architecture aligns with the research hypothesis. It has separate modules for maps, userspace code generation, eBPF C generation, kernel-module generation, context-specific code generation, safety checking, BTF parsing, dynptr bridging, tail-call analysis, and `struct_ops` registry logic. These modules encode eBPF concepts directly rather than treating the language as a thin preprocessor over C. The design therefore provides places where domain-specific checks can be added: map type compatibility, context field typing, stack estimates, function signature validation, and userspace API generation.

6 Evaluation Methodology

We evaluate KernelScript as a compiler and systems artifact. The goal is to test the scientific claim that the unified-source model reduces development complexity while preserving a path to ordinary eBPF artifacts. We do not claim packet-rate performance equivalence in this paper, because the public repository does not include matched hand-written C/libbpf baselines or a traffic-generation harness. Instead, we focus on reproducible evidence that the language covers representative domains, rejects some errors early, and generates buildable artifacts.

The evaluation runs on Linux 6.15.11-061511-generic using `clang 18`, `bpftool 7.7`, `libbpf 1.3.0`, `OCaml 4.14`, and `dune 3.14`. The evaluated commit is `6f9e6e8`. The artifact contains two scripts. First, `experiments/run_evaluation.py` builds the compiler, runs `dune runtest --force`, compiles every `examples/*.ks` file, runs `make` in each generated project, and records CSV/JSON results. Second, `experiments/run_smoke.sh` compiles a minimal loopback XDP program and uses `sudo -n` to run the generated loader. Third, `experiments/run_microbench.py` compares two KernelScript-generated XDP objects with equivalent hand-written C/eBPF objects using `bpftool`’s `BPF_PROG_TEST_RUN` path.

We measure four outcomes. **Correctness of the implementation** is approximated by the repository test suite. **Expressiveness** is approximated by whether examples for XDP, TC, probes, tracepoints, perf events, maps, ring buffers, dynptrs, tail calls, kfuncs, and `struct_ops` pass through the compiler and

generated build. **Complexity reduction** is measured as an expansion factor: lines of KernelScript source compared to generated C and `Makefile` lines. **Compatibility** is measured by generated build successes and failures against the local kernel toolchain.

The harness treats generated data as first-class evidence. It writes raw `stdout` and `stderr` logs for every compiler and `make` invocation, CSV rows for each example, JSON summaries for scripts, and a small TeX file containing the numeric macros used in this paper. This organization is deliberately close to artifact-evaluation practice: the paper does not rely on a spreadsheet that was manually edited after the run, and the reader can trace each aggregate number to a row in `results/examples_summary.csv`. We also keep temporary build products under `results/build` so that failures can be inspected after the harness exits.

SLOC is counted as nonblank, noncomment lines. Generated `vmlinux.h` and generated skeleton headers are excluded from the expansion factor because they are `bpftool` outputs rather than KernelScript compiler outputs. The expansion therefore counts only source and build logic produced directly by KernelScript: userspace C, eBPF C, optional kernel-module C, and `Makefile` lines. This definition is conservative for the generated-structure claim because it omits the large skeleton header that a low-level build still needs.

7 Results

Test suite. The compiler test suite reports 85 successful suites and 1092 passing tests with no reported failures. The tests cover lexer/parser behavior, type checking, maps, map assignment, userspace code generation, context field typing, safety checking, ring buffers, perf event attach, detach APIs, kfunc attributes, `struct_ops`, BTF parsing, and other regression cases. This breadth is important because the evaluation of a DSL depends on whether the implementation actually encodes domain concepts rather than only accepting example syntax.

Example buildability. Table 1 summarizes the example build results. Of 44 examples, 43 compile from KernelScript source. 41 then build into generated C/eBPF artifacts. The single compile-time rejection is `safety_demo.ks`, where safety analysis detects 608 bytes of stack usage and halts before C generation. This is a positive result for the safety hypothesis: the compiler catches a verifier-relevant problem early and provides a specific repair suggestion.

The two generated build failures are `sched_ext_simple.ks` and `struct_ops_simple.ks`. Both compile to eBPF objects and generate skeletons, but the generated userspace skeleton code references a `struct bpf_map_skeleton.link` field not available in the installed `libbpf 1.3.0` headers. This is a compatibility failure in the fast-moving `struct_ops` path, not a parser or type-checking failure. It is also a useful systems result: a unified-source language still inherits version skew across `bpftool`, `libbpf`, and kernel headers.

Table 1: Repository examples at commit 6f9e6e8.

Outcome	Count	Fraction
Examples total	44	100%
KernelScript compile success	43	97.7%
Generated project build success	41	93.2%
Safety rejection	1	2.3%
Struct_ops/libbpf mismatch	2	4.5%

Table 2: Feature coverage in repository examples. A = attempted, KS = KernelScript compiler success, B = generated project build success.

Feature	A	KS	B	Feature	A	KS	B
XDP	38	37	37	TC	5	5	5
Probe	4	4	4	Tracepoint	2	2	2
Perf	2	2	2	Maps	17	16	16
Ringbuf	3	3	3	Dynptr	1	1	1
kfunc	4	4	3	Struct_ops	2	2	0
Tail-call	2	2	2	Userspace	39	39	37

Feature coverage. The examples exercise many eBPF domains. Table 2 reports attempted examples, examples that pass the KernelScript compiler, and examples that fully build against the local toolchain. These counts should not be read as a complete language coverage proof; they show that the artifact is broader than a tracing-only DSL while making the struct_ops compatibility gap explicit.

The coverage table also highlights a weakness in the current benchmark suite: XDP dominates. That skew is common in early eBPF tools because XDP examples are easy to compile and do not require tracepoint discovery or workload generation. A later performance study should rebalance the corpus so that TC, perf events, ring buffers, kfuncs, and struct_ops each have multiple matched programs with comparable hand-written baselines.

Generated structure. For successful examples, the median KernelScript source size is 31 nonblank noncomment lines. The median generated project size, counting userspace C, eBPF C, optional module C, and Makefile lines but excluding generated `vmlinux.h` and skeleton headers, is 472 lines. The median expansion factor is 11.3x. This is evidence that a small unified source file centralizes hundreds of lines of low-level build and coordination code.

This metric is intentionally conservative. It does not say that every generated line is exactly a hand-written line saved, and it does not compare against an expert-written minimal C baseline. It does show that KernelScript centralizes recurring eBPF project structure. Examples such as perf events have particularly high expansion factors because attaching and reading counters requires substantial userspace coordination code relative to the kernel program itself.

Runtime microbenchmarks. Table 3 compares two generated XDP objects with hand-written C/eBPF baselines. Each program is loaded with `bpftool` and executed through `BPF_PROG_TEST_RUN` for 100000 repetitions over seven

Table 3: XDP BPF_PROG_TEST_RUN microbenchmarks.

Benchmark	Impl.	Instr.	Median ns (range)
pass	KernelScript	2	5 (5–6)
pass	C/eBPF	2	5 (5–6)
count	KernelScript	21	13 (12–13)
count	C/eBPF	11	9 (8–9)

trials; the table reports median average nanoseconds with the observed min–max range. The minimal pass program has the same median in this harness: 2 instructions and 5 (5–6) for KernelScript versus 5 (5–6) for C. The map counter is different. KernelScript emits a lookup followed by an update helper, while the C baseline uses an in-place atomic add on the array-map value. The generated version therefore has 21 instructions and 13 (12–13) median average time versus 11 instructions and 9 (8–9) for C. This result is small in absolute terms but important scientifically: the unified model need not impose overhead for trivial programs, but map assignment lowering can miss an optimization that an expert C programmer would write manually.

Generated code size. Among successful repository examples where disassembly is available, the median generated eBPF object contains 18.5 instructions. Small examples compile to tiny programs, while map-heavy examples generate much larger objects. This metric helps choose where runtime work is needed: examples with many generated map operations are the most likely to diverge from hand-written C.

Execution smoke test. The separate smoke test compiles `smoke_lo.ks`, builds the generated project, and runs the generated loader with `sudo`. The loader successfully attaches an XDP pass program to the loopback interface and detaches it: `XDP attached to interface: lo` followed by `XDP detached from interface index: 1`. This test is not a benchmark, but it validates that generated userspace code can complete a real load/attach/detach lifecycle on the test host.

8 Discussion

The evaluation supports KernelScript’s central research direction, but it also clarifies what remains unproven. The strongest supported claim is a complexity claim: one source model can generate the multi-file project structure needed for a variety of eBPF programs. The test suite and safety demo support a narrower safety claim: some verifier-relevant errors can be detected before kernel loading. The results establish parity only for a minimal pass program; they do not yet establish broad runtime performance parity with hand-written C/libbpf.

The struct_ops failures are instructive. A DSL can hide coordination code, but it cannot hide all ABI and API drift. When `bpftool`-generated skeletons and installed libbpf headers disagree, the generated userspace C fails. A production compiler would need version-aware generation, dependency

checks, or containerized toolchains. From a systems-research perspective, this is part of the contribution: the artifact exposes where a unified model reduces complexity and where the underlying Linux BPF stack still leaks through.

The expansion-factor metric should also be interpreted carefully. Generated C is not automatically better than hand-written C; it can be verbose, unidiomatic, or harder to debug. The practical benefit depends on diagnostic quality and source-level mapping from generated errors back to KernelScript source. Future work should therefore measure not only lines of code but time to fix injected errors, quality of compiler messages, and whether developers can understand generated verifier failures.

9 Threats to Validity

No expert C baseline. The expansion factor compares KernelScript source against generated artifacts, not against carefully hand-written C/libbpf programs. This is appropriate for measuring how much project structure the compiler creates, but it is not a direct developer-effort study. An expert could write smaller C for some examples, especially tiny XDP pass/drop programs. Conversely, generated code omits comments and may be denser than maintainable production C. The right interpretation is therefore “generated project structure centralized by the compiler,” not “exact lines saved.”

Repository examples may be biased. The corpus comes from the public KernelScript repository, so it naturally reflects the features the implementation authors wanted to demonstrate. It is useful for an artifact study because it is reproducible and covers many features, but it is not a representative sample of all eBPF applications. A stronger corpus would include programs from Cilium, bcc/libbpf-tools, production XDP filters, scheduler extensions, and observability agents, then reimplement each in KernelScript and C/libbpf.

Build success is not verifier success for every runtime path. The harness compiles eBPF objects and links userspace loaders. The smoke test loads and attaches one XDP program, but most repository examples are not executed because they attach to fixed devices, kernel functions, or performance events that may perturb the host. Thus, buildability is evidence of toolchain compatibility, not full deployment validation. A complete deployment study would run each example in a VM with known kernel configuration and explicit cleanup checks after detach.

Kernel version skew matters. The struct_ops failures show that results depend on the alignment between kernel headers, bpftool, generated skeleton code, and libbpf headers. This threat is not incidental: eBPF programming is tightly coupled to kernel versions. A production KernelScript compiler should record the target kernel ABI, check libbpf feature support before generation, and fail early with a dependency diagnostic when a generated feature cannot be compiled locally.

10 Toward Broader Runtime Evaluation

The microbenchmarks should be extended into a matched runtime study. For XDP, the study should compare packet counters, drop/pass filters, and map-based rate limiters against hand-written C/libbpf versions using `xdp-bench` or `pkt-gen`. For TC, the benchmark should use network namespaces and veth pairs to avoid host-network disruption. For perf events and ring buffers, the benchmark should measure event throughput and userspace loss under controlled producer rates. For struct_ops, the study must first pin a compatible kernel/libbpf/bpftool stack so that failures are about KernelScript design rather than dependency skew.

11 Related Work

C/libbpf remains the reference workflow for general-purpose eBPF development [4]. It exposes the full API surface and integrates with BTF and CO-RE, but it requires developers to manage multiple artifacts manually. Go and Rust libraries such as cilium/ebpf and Aya improve userspace integration and type safety within their host languages, but still separate the eBPF program from the controlling application [1, 3]. bpfftrace takes the opposite approach: it offers a concise high-level language, but focuses on tracing rather than arbitrary XDP, TC, struct_ops, or kfunc-oriented applications [2].

KernelScript occupies a different point in the design space. It is compiled rather than interpreted, broader than tracing-only languages, and more opinionated than host-language bindings. Its closest research question is whether eBPF application structure can be made explicit enough for a compiler to generate the low-level project while preserving compatibility with the Linux BPF toolchain. The public KernelScript repository [5] is therefore best read as both an implementation and a concrete design artifact for this space.

12 Conclusion

KernelScript is an early but useful prototype for studying typed unified-source eBPF development. Our artifact evaluation shows that the current compiler passes a broad test suite, compiles most repository examples, builds most generated projects, rejects an eBPF stack-limit violation before C generation, and can run a minimal XDP attach/detach smoke test. The median successful example expands from 31 lines of KernelScript to 472 lines of generated C and build logic, giving concrete evidence that the model centralizes recurring project structure. The remaining gaps are equally clear: struct_ops compatibility needs version-aware handling, and runtime performance is mixed beyond trivial XDP pass programs. Taken together, the results frame KernelScript’s scientific contribution as an exploration of a safer, unified programming model for eBPF, with a reproducible path for evaluating whether

that model reduces complexity without abandoning the kernel toolchain.

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